IGCSE Mathematics 0580 – Paper 4 Calculator (Extended)

Revision Paper, October/November 2026 – **Worked Solutions**

How to use this document. For every question part you will see: (1) the question, (2) the teacher explanation, which **starts with the original formula in symbols**, then **substitutes the values**, then explains, (3) the mark scheme, then a **bold black line** and (4) a **green box headed WRITE IN THE ANSWER SPACE** with the exact answer to write, including the working that earns the method marks. The explanation and the mark scheme are separated by a thin line.

Mark scheme codes. M = method mark. A = accuracy mark (only given if the method is right). B = independent mark. P = plotting mark. C = curve mark. FT = follow through from your own earlier answer. Give answers to 3 significant figures, and angles to 1 decimal place, unless the question says otherwise. Never round part way through.

Total for the paper: 100 marks.

Question 1 – Solving a linear equation [3 marks]

QUESTION

Solve.

$$4(3x - 5) = 2(x + 9) + 2$$

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Expanding brackets: $a(bx + c) = abx + ac$

Solving: $ax = b \implies x = b \div a$

SUBSTITUTE THE VALUES

$4(3x - 5) = 12x - 20$ and $2(x + 9) + 2 = 2x + 20$

$12x - 20 = 2x + 20 \implies 10x = 40 \implies x = 40 \div 10 = 4$

Expand both brackets first. Then collect the x terms on one side and the numbers on the other: $12x - 2x = 20 + 20$, so $10x = 40$. Do not divide by 4 at the start – the +2 on the right is not a multiple of 4.

2. MARK SCHEME

Answer	$12x - 20 = 2x + 18 + 2$ [M1] $10x = 40$ [M1] $x = 4$ [A1]
Guidance	M1 for expanding both brackets correctly. M1 for collecting terms correctly (follow through their expansion). A1 for $x = 4$.
Marks	**3**

WRITE IN THE ANSWER SPACE

$12x - 20 = 2x + 18 + 2$

$10x = 40$

$\boldsymbol{x = 4}$

Question 2 – Calculator work and significant figures [2 marks]

QUESTION

Work out.

$$\frac{16.71 + 46.13}{\sqrt{8.6^2 - 3.5^2}}$$

Give your answer correct to 2 significant figures.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Value} = \frac{p + q}{\sqrt{(a^2 - b^2)}}$$

Round to s significant figures: count from the first non-zero digit

SUBSTITUTE THE VALUES

$p = 16.71$, $q = 46.13$, $a = 8.6$, $b = 3.5$, $s = 2$

$$\text{Value} = \frac{16.71 + 46.13}{\sqrt{(8.6^2 - 3.5^2)}} = \frac{62.84}{\sqrt{61.71}}$$

Work out the top: 16.71 + 46.13 = 62.84. Work out the bottom: $8.6^2 - 3.5^2 = 73.96 - 12.25 = 61.71$, and $\sqrt{61.71} = 7.85557\ldots$ Then $62.84 \div 7.85557\ldots = 7.9994\ldots$ To 2 significant figures this is **8.0**, not 8 – the zero is needed to show 2 significant figures. On the calculator put the whole numerator and the whole denominator in brackets.

2. MARK SCHEME

Answer	7.99… (7.9994…) seen [M1] 8.0 [A1]
Guidance	Answer 8 only scores M1, A0. The whole working must keep the unrounded value.
Marks	**2**

WRITE IN THE ANSWER SPACE

Top = 62.84

Bottom = $\sqrt{(8.6^2 - 3.5^2)} = \sqrt{61.71} = 7.85557\ldots$

$62.84 \div 7.85557\ldots = 7.9994\ldots$

8.0

Question 3 – Reverse percentages [2 marks]

QUESTION

In a sale, the price of a printer is reduced by 18% to $287.
Work out the original price of the printer.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Reverse percentage: original $\times \frac{100 - p}{100}$ = sale price

$\Longrightarrow$ original = sale price $\div \frac{100 - p}{100}$ (p = percentage reduction)

SUBSTITUTE THE VALUES

$p = 18$, sale price = 287

original = $287 \div \frac{100 - 18}{100} = 287 \div 0.82$

After an 18% reduction the sale price is **82%** of the original. Going back you **divide**: 287 ÷ 0.82 = 350. Check: 350 × 0.82 = 287. Do not find 18% of 287 and add it on – that gives $338.66 and scores nothing.

2. MARK SCHEME

Answer	287 ÷ 0.82 (or 287 ÷ 82 × 100) [M1] 350 [A1]
Guidance	Answer 338.66 scores 0.
Marks	**2**

WRITE IN THE ANSWER SPACE

100% − 18% = 82%

Original price = 287 ÷ 0.82

$350

Question 4 – Sharing in a ratio [2 marks]

QUESTION

Amira, Bilal and Chen share $4560 in the ratio 5 : 7 : 12.
Calculate the amount Chen receives.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

One part = total ÷ (sum of the ratio numbers)

Share = (that person's ratio number) × one part

SUBSTITUTE THE VALUES

One part = 4560 ÷ (5 + 7 + 12) = 4560 ÷ 24 = 190

Chen's share = 12 × 190

Add the parts: 5 + 7 + 12 = 24. **Find one part first**: 4560 ÷ 24 = 190. Chen has 12 parts: 12 × 190 = 2280. Check: 950 + 1330 + 2280 = 4560.

2. MARK SCHEME

Answer	4560 ÷ 24 (or 190 seen) [M1] 2280 [A1]
Guidance	Accept 12/24 × 4560 for M1.
Marks	**2**

WRITE IN THE ANSWER SPACE

5 + 7 + 12 = 24

1 part = 4560 ÷ 24 = 190

Chen = 12 × 190

$2280

Question 5 – Sequences [3 marks]

QUESTION 5 – MAIN PART

These are the first 5 terms of a sequence.

2 6 12 20 30

5(a) [1 mark]

QUESTION

(a) Find the next term of this sequence.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Next term = last term + next first difference

The first differences go up by the second difference each time

SUBSTITUTE THE VALUES

First differences: 4, 6, 8, 10 (second difference = 2), so the next one is 10 + 2 = 12

Next term = 30 + 12

Find the differences: 4, 6, 8, 10. They go up by 2 each time, so the next difference is 12. Next term = 30 + 12 = 42.

2. MARK SCHEME

Answer	42 [B1]
Marks	**1**

WRITE IN THE ANSWER SPACE

30 + 12

42

5(b) [2 marks]

QUESTION

(b) Find the nth term of this sequence.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Quadratic sequence: nth term = $an^2 + bn + c$

a = (second difference) ÷ 2

SUBSTITUTE THE VALUES

$a = 2 \div 2 = 1$, so subtract n^2: 2, 6, 12, 20, 30 − (1, 4, 9, 16, 25) = 1, 2, 3, 4, 5

1, 2, 3, 4, 5 = n, so $b = 1$ and $c = 0$: nth term = $n^2 + n$

The first differences (4, 6, 8, 10) are not constant, but the **second differences** are constant (2, 2, 2). So the sequence is quadratic. Subtract n^2 (1, 4, 9, 16, 25) from the terms: 1, 2, 3, 4, 5, which is n. So the nth term is $n^2 + n$. Test it: $n = 4$ gives 16 + 4 = 20 ✓.

2. MARK SCHEME

Answer	$n^2 + n$ [2]
Guidance	M1 for n^2 seen, or for second difference 2. Accept $n(n + 1)$.
Marks	**2**

WRITE IN THE ANSWER SPACE

Second differences = 2, 2, 2, so n^2

2, 6, 12, 20, 30 − (1, 4, 9, 16, 25) = 1, 2, 3, 4, 5 = n

***n*th term = $n^2 + n$**

Question 6 – Factorising a quadratic [2 marks]

QUESTION

Factorise fully.

$$6x^2 - 7x - 3$$

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Factorising $ax^2 + bx + c$: find p and q with $p \times q = a \times c$ and $p + q = b$

then $ax^2 + px + qx + c$ and factorise in pairs

SUBSTITUTE THE VALUES

$a = 6$, $b = -7$, $c = -3$: $p \times q = 6 \times (-3) = -18$ and $p + q = -7$

$p = -9$, $q = +2$: $6x^2 - 9x + 2x - 3$

Find two numbers that multiply to −18 and add to −7: **−9 and +2**. Split the middle term and factorise in pairs: $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$. Expand your answer to check it.

2. MARK SCHEME

Answer	$(3x + 1)(2x - 3)$ [2]
Guidance	M1 for $6x^2 - 9x + 2x - 3$, or for $(3x \pm 1)(2x \pm 3)$, or $(6x \ldots)(x \ldots)$ with a correct pair of numbers.
Marks	**2**

WRITE IN THE ANSWER SPACE

$6x^2 - 9x + 2x - 3$

$3x(2x - 3) + 1(2x - 3)$

$(3x + 1)(2x - 3)$

Question 7 – Compound interest [3 marks]

QUESTION

Amira invests \$950 at a rate of 3.4% per year compound interest.
Calculate the value of her investment at the end of 6 years.
Give your answer correct to the nearest dollar.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Compound interest: $A = P(1 + \frac{r}{100})^n$

A = final amount, P = start (principal), r = rate % per year, n = number of years

SUBSTITUTE THE VALUES

$P = 950$, $r = 3.4$, $n = 6$

$A = 950 \times (1 + \frac{3.4}{100})^6 = 950 \times 1.034^6$

Compound interest is **one multiplier raised to a power**. A 3.4% increase gives a multiplier of 1.034, and 6 years means the power 6: $950 \times 1.034^6 = 950 \times 1.222146\ldots = 1161.04\ldots$ Round once, at the end, to the nearest dollar. Do not work year by year on rounded figures – you will drift by a dollar or two.

2. MARK SCHEME

Answer	1.034 used as the multiplier [M1] 950×1.034^6 [M1] 1161 [A1]
Guidance	M2 can be given directly for 950×1.034^6. Accept 1161.04 for M2 A0 only if the dollar rounding is missing.
Marks	**3**

WRITE IN THE ANSWER SPACE

Multiplier = 1.034

$950 \times 1.034^6 = 1161.04\ldots$

\$1161

Question 8 – Simple interest [3 marks]

QUESTION

Bilal invests \$1330 at a rate of r% per year **simple** interest.
At the end of 8 years the total value of his investment is \$1968.40.
Calculate the value of r.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Simple interest: $I = \dfrac{PrT}{100} \quad \Rightarrow \quad r = \dfrac{100I}{PT}$

$I = A - P$ (interest = final amount − start)

SUBSTITUTE THE VALUES

$P = 1330$, $T = 8$, $I = 1968.40 - 1330 = 638.40$

$$r = \frac{100 \times 638.40}{1330 \times 8} = \frac{63\,840}{10\,640} = 6$$

Subtract the starting amount to get the interest earned: 1968.40 − 1330 = \$638.40. Simple interest is the same every year: 638.40 ÷ 8 = \$79.80. As a percentage of the \$1330: 79.80 ÷ 1330 × 100 = 6. Do not divide 1968.40 by 1330 – that gives the total growth factor 1.48, not the yearly rate.

2. MARK SCHEME

Answer	1968.40 − 1330 = 638.40 [M1] 638.40 ÷ 8 = 79.80 [M1] $r = 6$ [A1]
Guidance	Alternative: 638.40 ÷ 1330 = 0.48 (48%) then ÷ 8 [M1 M1 A1].
Marks	**3**

WRITE IN THE ANSWER SPACE

Interest = 1968.40 − 1330 = 638.40

Per year = 638.40 ÷ 8 = 79.80

$$r = \frac{79.80}{1330} \times 100$$

$\boldsymbol{r = 6}$

Question 9 – Simultaneous equations [3 marks]

QUESTION

Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 4$
$5x - 4y = 25$

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Elimination: multiply an equation by k: $k(ax + by) = kc$

same signs → subtract the equations; opposite signs → add them

then substitute back into an original equation

SUBSTITUTE THE VALUES

× 2 on $3x + 2y = 4$: $6x + 4y = 8$

$+4y$ and $-4y$ have opposite signs, so add: $6x + 5x = 8 + 25 \implies 11x = 33$

Make the y coefficients match: the first equation has $+2y$ and the second has $-4y$, so **double the first equation**: $6x + 4y = 8$. The signs are now opposite ($+4y$ and $-4y$), so **add** the equations: $11x = 33$, so $x = 3$. Substitute into the first equation: $9 + 2y = 4$, $2y = -5$, $y = -2.5$. Check in the other equation: $5(3) - 4(-2.5) = 15 + 10 = 25$ ✓.

2. MARK SCHEME

Answer	$6x + 4y = 8$ (equal y coefficients, correct add/subtract) [M1] $x = 3$ [A1] $y = -2.5$ [A1]
Guidance	Accept any correct elimination or substitution method.
Marks	**3**

WRITE IN THE ANSWER SPACE

× 2: $6x + 4y = 8$
Add: $11x = 33$
$x = 3$
$9 + 2y = 4$, so $2y = -5$
***x* = 3, *y* = −2.5**

Question 10 – Simplifying an algebraic fraction [3 marks]

QUESTION

Simplify fully.

$$\frac{3x^2 - 12}{x^2 - x - 6}$$

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Difference of two squares: $a^2 - b^2 = (a - b)(a + b)$

Quadratic: $x^2 + (p + q)x + pq = (x + p)(x + q)$

Cancel a common factor: $\frac{F \times A}{F \times B} = \frac{A}{B}$

SUBSTITUTE THE VALUES

Top: $3x^2 - 12 = 3(x^2 - 4) = 3(x^2 - 2^2) = 3(x - 2)(x + 2)$

Bottom: $p + q = -1$ and $p \times q = -6$, so $p = -3$, $q = 2$: $(x - 3)(x + 2)$

Factorise the top and the bottom, then cancel. **Top:** take out 3 first, $3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$. **Bottom:** two numbers that multiply to −6 and add to −1 are −3 and +2, so $x^2 - x - 6 = (x - 3)(x + 2)$. Cancel the common factor $(x + 2)$. Never cancel across a + or − sign.

2. MARK SCHEME

Answer	$3(x - 2)(x + 2)$ [B1] $(x - 3)(x + 2)$ [B1] $\frac{3(x - 2)}{x - 3}$ [B1]
Guidance	Accept $\frac{3x - 6}{x - 3}$ for the final mark.
Marks	**3**

WRITE IN THE ANSWER SPACE

Top: $3(x^2 - 4) = 3(x - 2)(x + 2)$

Bottom: $(x - 3)(x + 2)$

Cancel $(x + 2)$

$$\mathbf{\frac{3(x - 2)}{x - 3}}$$

Question 11 – Changing the subject of a formula [4 marks]

QUESTION

Make x the subject of this formula.

$$y = \frac{3x + 2}{x - 5}$$

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

If $y = \frac{A}{B}$ then $y \times B = A$ (multiply out the fraction)

Collect every x term on one side, factorise: $x(\ldots) = \ldots$, then divide

SUBSTITUTE THE VALUES

$$y = \frac{3x + 2}{x - 5} \implies y(x - 5) = 3x + 2$$

$$xy - 5y = 3x + 2 \implies xy - 3x = 5y + 2 \implies x(y - 3) = 5y + 2$$

x is in the formula twice, so you must **collect both x terms and factorise**. Multiply both sides by $(x - 5)$, expand, move every x term to the left and everything else to the right, take x out as a common factor, then divide by $(y - 3)$.

2. MARK SCHEME

Answer	$y(x - 5) = 3x + 2$ [M1] $xy - 5y = 3x + 2$ [M1] $xy - 3x = 5y + 2$, then $x(y - 3) = 5y + 2$ [M1] $x = \frac{5y + 2}{y - 3}$ [A1]
Guidance	Accept equivalent forms, for example $\frac{-5y - 2}{3 - y}$.
Marks	**4**

WRITE IN THE ANSWER SPACE

$y(x - 5) = 3x + 2$

$xy - 5y = 3x + 2$

$xy - 3x = 5y + 2$

$x(y - 3) = 5y + 2$

$$\boldsymbol{x = \frac{5y + 2}{y - 3}}$$

Question 12 – Interior angle of a regular polygon [2 marks]

QUESTION

The interior angle of a regular polygon is 162°.
Find the number of sides of this polygon.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Exterior angle = 180° − interior angle

Number of sides n = 360° ÷ exterior angle

SUBSTITUTE THE VALUES

Exterior angle = 180° − 162° = 18°

n = 360° ÷ 18° = 20

Go through the **exterior angle** – it is the quicker route. The exterior angles of any polygon add up to 360°, so the number of sides = 360 ÷ 18 = 20.

2. MARK SCHEME

Answer	180 − 162 = 18 (or 360 ÷ 18 attempted) [M1] 20 [A1]
Guidance	Accept (n − 2) × 180 ÷ n = 162 leading to n = 20 for M1 A1.
Marks	**2**

WRITE IN THE ANSWER SPACE

Exterior angle = 180° − 162° = 18°

Number of sides = 360 ÷ 18

20 sides

Question 13 – Turning points [5 marks]

QUESTION

A curve has equation $y = x^3 - 6x^2 + 9x + 2$.
Find the coordinates of the two turning points of the curve and determine whether each one is a maximum or a minimum.
You must show all your working.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Differentiate: if $y = ax^n$ then $\frac{dy}{dx} = n \times a \times x^{n-1}$

Turning point: $\frac{dy}{dx} = 0$

Nature: $\frac{d^2y}{dx^2} > 0 \rightarrow$ minimum; $\frac{d^2y}{dx^2} < 0 \rightarrow$ maximum

SUBSTITUTE THE VALUES

$y = x^3 - 6x^2 + 9x + 2 \implies \frac{dy}{dx} = 3x^2 - 12x + 9 = 0$

$\frac{d^2y}{dx^2} = 6x - 12$: at $x = 1$ it is -6 (maximum); at $x = 3$ it is $+6$ (minimum)

Turning points have **zero gradient**, so set the derivative to 0: $3(x^2 - 4x + 3) = 0$, $3(x - 1)(x - 3) = 0$, so $x = 1$ or $x = 3$. Substitute into the **original equation** (not the derivative) to get the y values: $x = 1$ gives $1 - 6 + 9 + 2 = 6$ and $x = 3$ gives $27 - 54 + 27 + 2 = 2$. For the nature, differentiate again. Negative at $x = 1$ so a **maximum**; positive at $x = 3$ so a **minimum**. (A positive cubic rises, turns down, then rises again, so the first turning point is the maximum.)

2. MARK SCHEME

Answer	$3x^2 - 12x + 9$ [M1] set = 0 and solve [M1] $x = 1$ and $x = 3$ [A1] (1, 6) and (3, 2) [A1] (1, 6) maximum and (3, 2) minimum, with a reason [B1]
Guidance	Accept the second-derivative test, a gradient test either side, or the shape of the cubic as the reason.
Marks	**5**

WRITE IN THE ANSWER SPACE

$dy/dx = 3x^2 - 12x + 9$

$3x^2 - 12x + 9 = 0$

$3(x - 1)(x - 3) = 0$, so $x = 1$ or $x = 3$

$x = 1$: $y = 1 - 6 + 9 + 2 = 6$ $x = 3$: $y = 27 - 54 + 27 + 2 = 2$

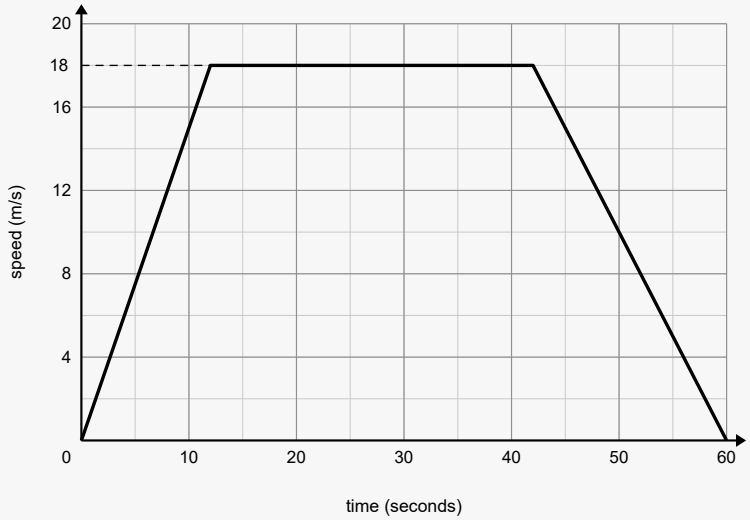
$d^2y/dx^2 = 6x - 12$: -6 at $x = 1$, $+6$ at $x = 3$

(1, 6) is a maximum (3, 2) is a minimum

Question 14 – Speed–time graph [5 marks]

QUESTION 14 – MAIN PART

The speed–time graph shows the journey of a car.



14(a) [1 mark]

QUESTION

(a) Calculate the acceleration during the first 12 seconds.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Acceleration} = \frac{\text{change in speed}}{\text{time taken}} = \frac{v - u}{t}$$ (the gradient of a speed–time graph)

SUBSTITUTE THE VALUES

$u = 0$, $v = 18$, $t = 12$: $\text{acceleration} = \frac{18 - 0}{12}$

Acceleration is the **gradient** of a speed–time graph. In the first 12 seconds the speed rises from 0 to 18 m/s, so (18 − 0) ÷ 12 = 1.5. The unit is m/s^2.

2. MARK SCHEME

Answer	1.5 [B1]
Guidance	Unit m/s^2 is not required for the mark.
Marks	**1**

WRITE IN THE ANSWER SPACE

$$\text{Acceleration} = \frac{18 - 0}{12}$$

1.5 m/s^2

14(b) [3 marks]

QUESTION

(b) Calculate the total distance travelled.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Distance = area under the speed–time graph

Area of a trapezium = ½(a + b) × h (a, b = parallel sides, h = distance between them)

SUBSTITUTE THE VALUES

a = 42 − 12 = 30, b = 60, h = 18

Area = ½(30 + 60) × 18

Distance is the **area under** a speed–time graph. The shape is a trapezium. The parallel sides are the top (from 12 s to 42 s, so 42 − 12 = **30**, not 42) and the bottom (60 s). The height is 18 m/s. Area = ½(30 + 60) × 18 = 810. Check by splitting: triangle 108 + rectangle 540 + triangle 162 = 810 ✓.

2. MARK SCHEME

Answer	Area under the graph attempted (trapezium, or triangle + rectangle + triangle) [M1] ½(30 + 60) × 18, or 108, 540, 162 [M1] 810 [A1]
Guidance	Using 42 instead of 30 for the top side loses the second M1.
Marks	**3**

WRITE IN THE ANSWER SPACE

Top side = 42 − 12 = 30 s

Distance = ½ × (30 + 60) × 18

= ½ × 90 × 18

810 m

14(c) [1 mark]

QUESTION

(c) Calculate the average speed for the whole journey.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$$

SUBSTITUTE THE VALUES

$$\text{Average speed} = \frac{810}{60}$$

Average speed = total distance ÷ total time = 810 ÷ 60 = 13.5 m/s.

2. MARK SCHEME

Answer	13.5 [B1]
Guidance	Follow through: their (b) ÷ 60.
Marks	**1**

WRITE IN THE ANSWER SPACE

$$\text{Average speed} = \frac{810}{60}$$

13.5 m/s

Question 15 – Exponential decay [4 marks]

QUESTION 15 – MAIN PART

A machine is worth \$24 000.
Its value decreases exponentially by 12% each year.

15(a) [2 marks]

QUESTION

(a) Calculate the value of the machine after 5 years.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Exponential decay: $V = V_0(1 - \frac{r}{100})^n$

V_0 = starting value, r = % lost each year, n = number of years

SUBSTITUTE THE VALUES

$V_0 = 24\,000$, $r = 12$, $n = 5$

$V = 24\,000 \times (1 - \frac{12}{100})^5 = 24\,000 \times 0.88^5$

Losing 12% leaves 88%, so the multiplier is **0.88**. After 5 years: $24\,000 \times 0.88^5 = 24\,000 \times 0.527731\ldots = 12\,665.57\ldots$, which is \$12 700 to 3 significant figures.

2. MARK SCHEME

Answer	$24\,000 \times 0.88^5$ [M1] 12 700 [A1]
Guidance	Accept 12 665.57 or 12 666. Using 1.12 instead of 0.88 scores 0.
Marks	**2**

WRITE IN THE ANSWER SPACE

Multiplier = 1 − 0.12 = 0.88

$24\,000 \times 0.88^5 = 12\,665.57\ldots$

\$12 700 (3 s.f.)

15(b) [2 marks]

QUESTION

(b) Find the number of complete years before the value of the machine first falls below $8000.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Same formula: $V = V_0 \times (\text{multiplier})^n$; find the first whole n with $V < 8000$

SUBSTITUTE THE VALUES

$24\,000 \times 0.88^n < 8000$: try $n = 8$ and $n = 9$

(with logs: $n > \dfrac{\log(8000 \div 24\,000)}{\log 0.88}$)

Try successive powers of 0.88 until the value drops below $8000. After 8 years: $24\,000 \times 0.88^8 = \$8631$, still above $8000. After 9 years: $24\,000 \times 0.88^9 = \$7595$, now below $8000. **Show both values** – the 9-year value alone does not prove that 8 years was still above $8000.

2. MARK SCHEME

Answer	$24\,000 \times 0.88^n$ tried for different values of n [M1] 9 [A1]
Guidance	Both 8631 (n = 8) and 7595 (n = 9) must be seen for full marks.
Marks	**2**

WRITE IN THE ANSWER SPACE

$24\,000 \times 0.88^8 = 8631$ (more than 8000)

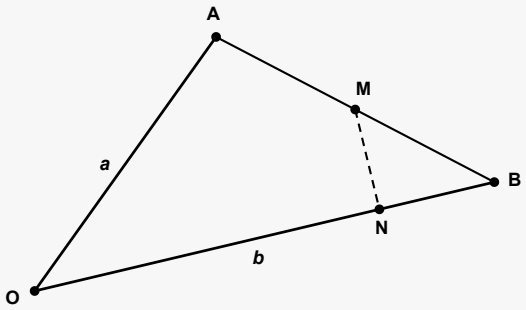
$24\,000 \times 0.88^9 = 7595$ (less than 8000)

9 years

Question 16 – Vectors [5 marks]

QUESTION 16 – MAIN PART

In the diagram, **OA** = ***a*** and **OB** = ***b***.
M is the midpoint of AB and N is the point on OB such that ON : NB = 3 : 1.
Find, in terms of ***a*** and ***b***, giving each answer in its simplest form.



16(a) [1 mark]

QUESTION

(a) AB

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Vector from X to Y: **XY** = −**OX** + **OY** = **OY** − **OX**

SUBSTITUTE THE VALUES

AB = −**OA** + **OB** = −***a*** + ***b***

To go from A to B, go back to O and then forward to B: **AB** = −**OA** + **OB** = −***a*** + ***b*** = ***b*** − ***a***. One rule solves every vector question.

2. MARK SCHEME

Answer	***b*** − ***a*** [B1]
Guidance	Accept −***a*** + ***b***.
Marks	**1**

WRITE IN THE ANSWER SPACE

AB = −***a*** + ***b***

AB = *b* − *a*

16(b) [2 marks]

QUESTION

(b) OM

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Midpoint: **OM** = **OA** + ½**AB**

SUBSTITUTE THE VALUES

OM = ***a*** + ½(***b*** − ***a***)

M is the midpoint of AB, so **OM** = ***a*** + ½(***b*** − ***a***) = ***a*** − ½***a*** + ½***b*** = ½***a*** + ½***b***.

2. MARK SCHEME

Answer	***a*** + ½(***b*** − ***a***) [M1] ½(***a*** + ***b***) [A1]
Guidance	Accept ½***a*** + ½***b***.
Marks	**2**

WRITE IN THE ANSWER SPACE

OM = ***a*** + ½(***b*** − ***a***)

= ***a*** − ½***a*** + ½***b***

OM = ½(*a* + *b*)

16(c) [2 marks]

QUESTION

(c) MN

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Ratio: if ON : NB = m : n then $\mathbf{ON} = \frac{m}{m+n}\,\mathbf{OB}$

$\mathbf{MN} = -\mathbf{OM} + \mathbf{ON}$

SUBSTITUTE THE VALUES

$m = 3$, $n = 1$: $\mathbf{ON} = \frac{3}{3+1}\boldsymbol{b} = \frac{3}{4}\boldsymbol{b}$

$\mathbf{MN} = -\frac{1}{2}(\boldsymbol{a} + \boldsymbol{b}) + \frac{3}{4}\boldsymbol{b}$

ON : NB = 3 : 1 means there are 3 + 1 = **4 parts** (quarters, not thirds), and N is $\frac{3}{4}$ of the way along OB, so $\mathbf{ON} = \frac{3}{4}\boldsymbol{b}$. Then $\mathbf{MN} = -\mathbf{OM} + \mathbf{ON} = -\frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b} + \frac{3}{4}\boldsymbol{b} = \frac{1}{4}\boldsymbol{b} - \frac{1}{2}\boldsymbol{a}$.

2. MARK SCHEME

Answer	$\mathbf{ON} = \frac{3}{4}\boldsymbol{b}$ and $\mathbf{MN} = -\mathbf{OM} + \mathbf{ON}$ [M1] $\frac{1}{4}\boldsymbol{b} - \frac{1}{2}\boldsymbol{a}$ [A1]
Guidance	Accept $\frac{1}{2}(\frac{1}{2}\boldsymbol{b} - \boldsymbol{a})$.
Marks	**2**

WRITE IN THE ANSWER SPACE

$\mathbf{ON} = \frac{3}{4}\boldsymbol{b}$

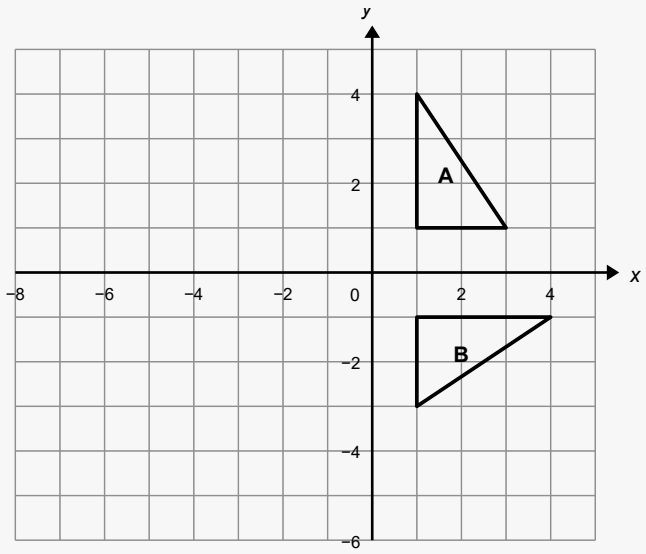
$\mathbf{MN} = -\mathbf{OM} + \mathbf{ON} = -\frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b} + \frac{3}{4}\boldsymbol{b}$

$\mathbf{MN} = \frac{1}{4}\boldsymbol{b} - \frac{1}{2}\boldsymbol{a}$

Question 17 – Transformations [5 marks]

QUESTION 17 – MAIN PART

The diagram shows triangle A and triangle B on a grid.



17(a) [3 marks]

QUESTION

(a) Describe fully the **single** transformation that maps triangle A onto triangle B.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Rotation 90° clockwise about the origin: $(x, y) \rightarrow (y, -x)$

Rotation 90° anticlockwise about the origin: $(x, y) \rightarrow (-y, x)$

Describe fully: **rotation** + **angle and direction** + **centre**

SUBSTITUTE THE VALUES

$(1, 1) \rightarrow (1, -1)$ $(3, 1) \rightarrow (1, -3)$ $(1, 4) \rightarrow (4, -1)$

All three fit $(x, y) \rightarrow (y, -x)$: rotation, 90° clockwise, centre (0, 0)

B is the same size and shape as A but turned, so the transformation is a **rotation**. Only the origin does not move, so the centre is (0, 0). “Describe fully” needs **three facts**: the word, the angle with its direction, and the centre. Naming two transformations scores zero, because the question says *single*.

2. MARK SCHEME

Answer	rotation [B1] 90° clockwise [B1] centre (0, 0) [B1]
Guidance	Accept 270° anticlockwise. Accept origin for (0, 0).
Marks	**3**

WRITE IN THE ANSWER SPACE

Rotation, 90° clockwise, centre (0, 0)

17(b) [2 marks]

QUESTION

(b) On the grid, draw the image of triangle A after an enlargement with scale factor −2 and centre (0, 1).

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Enlargement, scale factor k, centre **C**: image = **C** + k × (**P** − **C**)

(a negative k puts the image on the opposite side of the centre)

SUBSTITUTE THE VALUES

C = (0, 1), k = −2

(1, 1): (0, 1) + (−2)(1, 0) = (−2, 1) (3, 1): (0, 1) + (−2)(3, 0) = (−6, 1) (1, 4): (0, 1) + (−2)(1, 3) = (−2, −5)

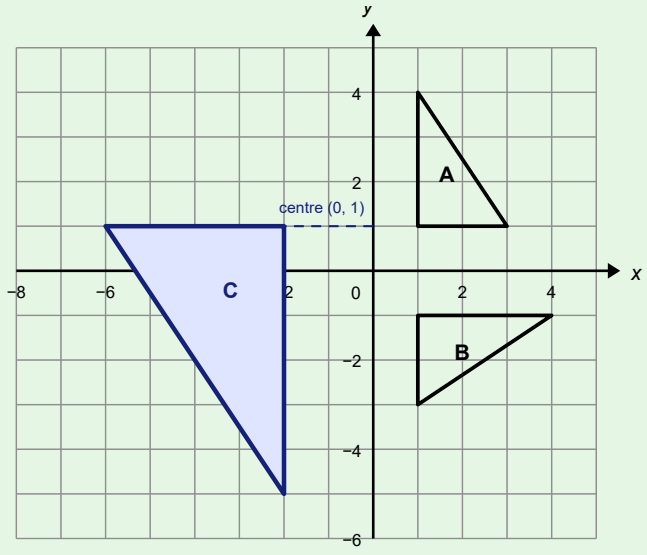
Measure from the centre (0, 1) to each vertex, multiply by −2, and go to the **opposite** side of the centre. The image is twice as large and upside down.

2. MARK SCHEME

Answer	triangle with vertices (−2, 1), (−6, 1) and (−2, −5) [2]
Guidance	B1 for a triangle of the correct size and orientation (twice as large, turned round) but in the wrong position.
Marks	**2**

WRITE IN THE ANSWER SPACE

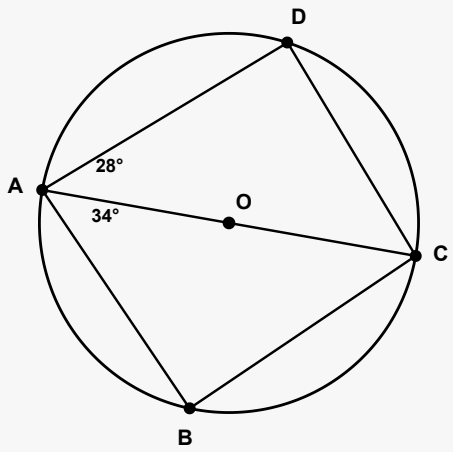
Vertices: (−2, 1), (−6, 1), (−2, −5). Draw and label the triangle C:



Question 18 – Circle theorems [5 marks]

QUESTION 18 – MAIN PART

A, B, C and D lie on a circle, centre O.
AC is a diameter, angle BAC = 34° and angle CAD = 28°.



Find the following angles, giving a geometrical reason for each answer.

18(a) [2 marks]

QUESTION

(a) Angle ABC, with a reason.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

The angle in a semicircle = 90° (a diameter and a point on the circle)

SUBSTITUTE THE VALUES

AC is a diameter and B is on the circle, so angle ABC = 90°

B is on the circle and AC is a diameter, so angle ABC is the **angle in a semicircle**, which is always 90°. You must **name the theorem** – "because AC is a diameter" does not score the reason mark.

2. MARK SCHEME

Answer	90° [B1] angle in a semicircle (is a right angle) [B1]
Marks	**2**

WRITE IN THE ANSWER SPACE

Angle ABC = 90°
Reason: the angle in a semicircle is a right angle.

18(b) [1 mark]

QUESTION

(b) Angle BCA.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Angles in a triangle: $A + B + C = 180°$

SUBSTITUTE THE VALUES

Angle BCA = 180° − 90° − 34° = 56°

Use triangle ABC. The angles of a triangle add up to 180°: 180 − 90 − 34 = 56°.

2. MARK SCHEME

Answer	56° [B1]
Guidance	No reason is required for this part.
Marks	**1**

WRITE IN THE ANSWER SPACE

180 − 90 − 34

Angle BCA = 56°

18(c) [2 marks]

QUESTION

(c) Angle BCD, with a reason.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Opposite angles of a cyclic quadrilateral: $A + C = 180°$

SUBSTITUTE THE VALUES

Angle BAD = 34° + 28° = 62°

Angle BCD = 180° − 62° = 118°

First find the whole angle at A: angle BAD = 34° + 28° = **62°** (the question needs BAD, not BAC). ABCD is a **cyclic quadrilateral** and A and C are opposite corners, so they add up to 180°: 180 − 62 = 118°.

2. MARK SCHEME

Answer	118° [B1] opposite angles of a cyclic quadrilateral add up to 180° [B1]
Marks	**2**

WRITE IN THE ANSWER SPACE

Angle BAD = 34° + 28° = 62°

Angle BCD = 180° − 62° = 118°

Reason: opposite angles in a cyclic quadrilateral add up to 180°.

Question 19 – Cubic graph and tangent [9 marks]

QUESTION 19 – MAIN PART

The table shows some values for $y = x^3 - 3x - 1$.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2	2.5
y	−3		1	0.375	−1	−2.375	−3		1	7.125

The grid for part (b) is printed in the question paper.

19(a) [2 marks]

QUESTION

(a) Complete the table.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Substitute each x value into $y = x^3 - 3x - 1$

SUBSTITUTE THE VALUES

$x = -1.5$: $y = (-1.5)^3 - 3(-1.5) - 1 = -3.375 + 4.5 - 1 = 0.125$

$x = 1.5$: $y = (1.5)^3 - 3(1.5) - 1 = 3.375 - 4.5 - 1 = -2.125$

Use **brackets** on the calculator: $(-1.5)^3 = -3.375$, not $+3.375$.

2. MARK SCHEME

Answer	0.125 [B1] −2.125 [B1]
Marks	**2**

WRITE IN THE ANSWER SPACE

$x = -1.5$: $(-1.5)^3 - 3(-1.5) - 1 = 0.125$

$x = 1.5$: $(1.5)^3 - 3(1.5) - 1 = -2.125$

0.125 and −2.125

19(b) [4 marks]

QUESTION

(b) On the grid, draw the graph of $y = x^3 - 3x - 1$ for $-2 \leqslant x \leqslant 2.5$.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Plot the points (x, y) from the table, then join them with one smooth curve

SUBSTITUTE THE VALUES

(−2, −3) (−1.5, 0.125) (−1, 1) (−0.5, 0.375) (0, −1) (0.5, −2.375) (1, −3) (1.5, −2.125) (2, 1) (2.5, 7.125)

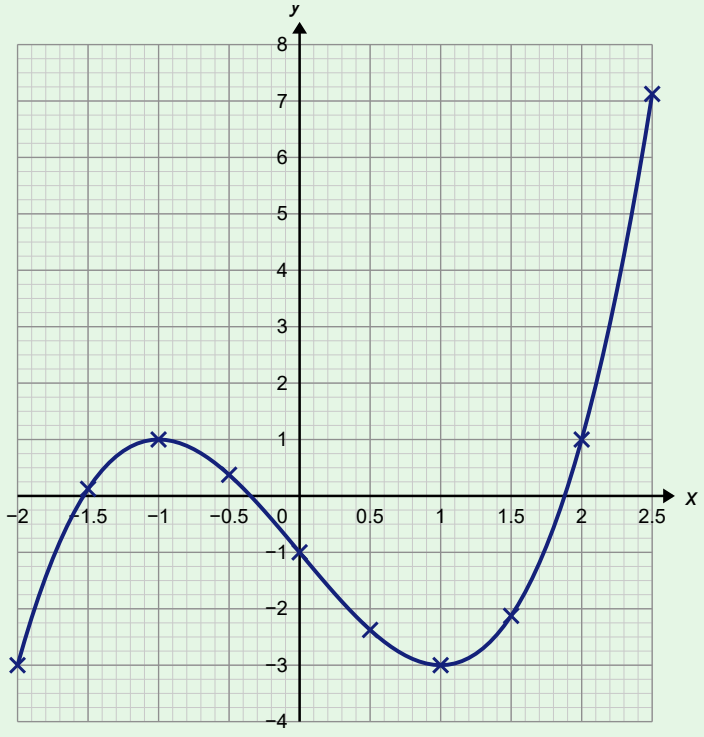
Plot all ten points carefully, with small crosses. Then join them with **one smooth curve**, drawn freehand with a sharp pencil – never with a ruler. The curve turns near (−1, 1) and (1, −3).

2. MARK SCHEME

Answer	all 10 points plotted correctly [P3] (P2 for 8 or 9 points, P1 for 5 to 7) smooth curve through the points [C1]
Guidance	Points must be within half a small square. No ruled lines between points.
Marks	**4**

WRITE IN THE ANSWER SPACE

Plot the ten points and draw the smooth curve:



19(c) [3 marks]

QUESTION

(c) By drawing a suitable tangent, estimate the gradient of the graph at the point where $x = 2$.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{(measured along the tangent)}$$

SUBSTITUTE THE VALUES

$(x_1, y_1) = (1.5, -3.5)$ and $(x_2, y_2) = (2.5, 5.5)$

$$\text{Gradient} = \frac{5.5 - (-3.5)}{2.5 - 1.5} = \frac{9}{1}$$

A tangent is a straight line that **just touches** the curve without cutting it. Put a ruler on the curve at (2, 1) and draw the tangent. Then take two points **far apart on your own tangent**. Anything from about 7.5 to 10.5 is accepted because it depends on your drawing. Calculus gives exactly $3(2)^2 - 3 = 9$, but quoting 9 with **no tangent drawn** scores no method marks.

2. MARK SCHEME

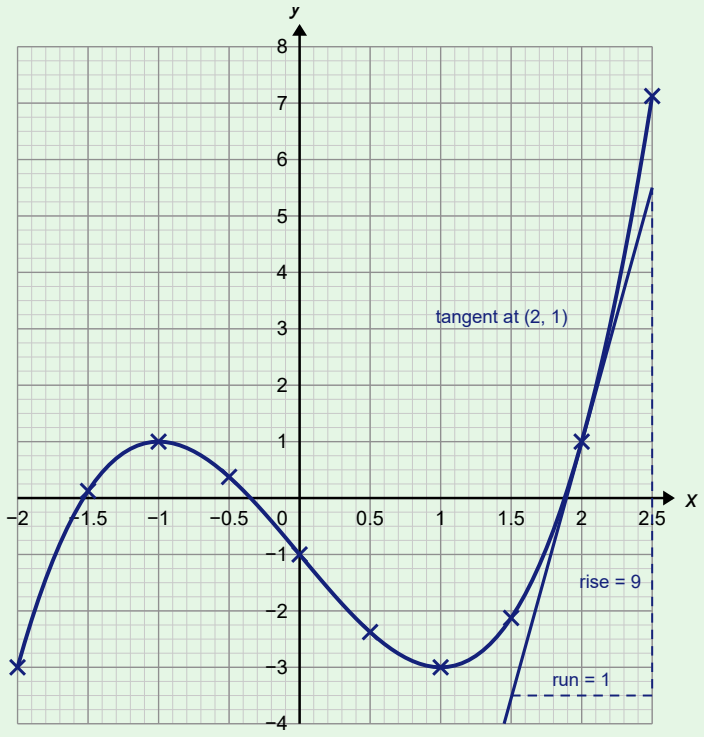
Answer	tangent drawn at (2, 1) [M1] gradient triangle or rise ÷ run from two points on the tangent [M1] 7.5 to 10.5 [A1]
Guidance	Follow through only from a tangent that has been drawn.
Marks	**3**

WRITE IN THE ANSWER SPACE

Tangent drawn at (2, 1). Rise = 5.5 − (−3.5) = 9, run = 2.5 − 1.5 = 1

Gradient = rise ÷ run = 9 ÷ 1

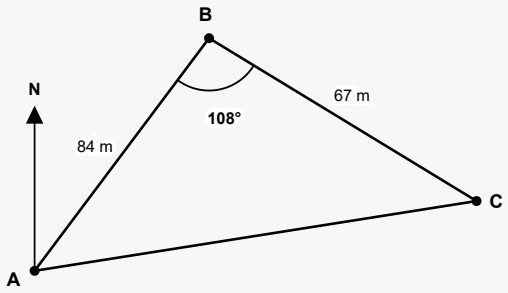
gradient ≈ 9



Question 20 – Cosine rule, sine rule and bearings [10 marks]

QUESTION 20 – MAIN PART

A, B and C are three points on level ground.
AB = 84 m, BC = 67 m and angle ABC = 108°.



20(a) [4 marks]

QUESTION

(a) Calculate AC.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$

(A is the angle between sides b and c; a is the side opposite A)

SUBSTITUTE THE VALUES

a = AC, $b = 84$, $c = 67$, $A = 108°$

$AC^2 = 84^2 + 67^2 - 2 \times 84 \times 67 \times \cos 108°$

You know two sides and the angle **between** them, so use the **cosine rule**. Check the calculator is in **DEG**. cos 108° = −0.309017… is negative, so the $-2bc\cos A$ term *adds*: 7056 + 4489 + 3478.3… = 15 023.3… Then AC = √15 023.3… = 122.5696… Keep this full value in the calculator for parts (b) and (d).

2. MARK SCHEME

Answer	$84^2 + 67^2 - 2 \times 84 \times 67 \times \cos 108°$ [M1] 15 023.3… (7056 + 4489 + 3478.3…) [M1] square root [M1] 123 (122.5696…) [A1]
Guidance	Accept 122.6 or 122.57. In radians the answer is 129.9, which scores 0. If AC comes out smaller than 84 the minus sign was dropped.
Marks	**4**

WRITE IN THE ANSWER SPACE

$AC^2 = 84^2 + 67^2 - 2(84)(67)\cos 108°$

$= 7056 + 4489 + 3478.3… = 15\,023.3…$

$AC = 122.5696…$

AC = 123 m (3 s.f.)

20(b) [3 marks]

QUESTION

(b) Calculate angle BAC.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Sine rule: $\frac{\sin A}{a} = \frac{\sin B}{b}$ (each side is opposite its own angle)

SUBSTITUTE THE VALUES

BC = 67 is opposite angle BAC; AC = 122.5696… is opposite angle B = 108°

$$\frac{\sin \text{BAC}}{67} = \frac{\sin 108°}{122.5696…}$$

Now you know a side and its opposite angle, so use the **sine rule** with the **unrounded** AC. Rearrange: sin BAC = 67 × sin 108° ÷ 122.5696… = 0.51987… and BAC = $\sin^{-1}(0.51987…)$ = 31.3°.

2. MARK SCHEME

Answer	sin BAC ÷ 67 = sin 108° ÷ 122.5696… [M1] sin BAC = 0.5198… [M1] 31.3 [A1]
Guidance	Accept 31.33° or 31°. The sine rule must have BC opposite angle A.
Marks	**3**

WRITE IN THE ANSWER SPACE

$$\frac{\sin \text{BAC}}{67} = \frac{\sin 108°}{122.5696…}$$

$$\sin \text{BAC} = \frac{67 \times \sin 108°}{122.5696…} = 0.51987…$$

Angle BAC = 31.3°

20(c) [1 mark]

QUESTION

(c) The bearing of B from A is 052° and C lies to the east of the line AB. Find the bearing of C from A.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

A bearing is measured clockwise from north, written with 3 figures

Bearing of C from A = bearing of B from A + angle BAC (C is clockwise of B)

SUBSTITUTE THE VALUES

Bearing of C from A = 052° + 31.3° = 083.3°

C is east of AB, so C is further round, clockwise, than B: 52° + 31.3° = 83.3°. Written with three figures: 083.3°.

2. MARK SCHEME

Answer	083.3 [B1]
Guidance	Accept 83.3. Follow through: 052 + their (b).
Marks	**1**

WRITE IN THE ANSWER SPACE

Bearing = 052° + 31.3°

083.3°

20(d) [2 marks]

QUESTION

(d) A vertical mast of height 30 m stands at C. Calculate the angle of elevation of the top of the mast from A.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

(angle of elevation: opposite = height, adjacent = horizontal distance)

SUBSTITUTE THE VALUES

opposite = 30, adjacent = AC = 122.5696…

$$\tan\theta = \frac{30}{122.5696\ldots}$$

The mast is vertical, so the triangle is right-angled at C. The horizontal distance is **AC** (not 84 and not 67). $\tan\theta = 30 \div 122.5696\ldots = 0.24476\ldots$ and $\theta = \tan^{-1}(0.24476\ldots) = 13.8°$.

2. MARK SCHEME

Answer	$\tan\theta = 30 \div 122.5696\ldots$ [M1] 13.8 [A1]
Guidance	Accept 13.7° or 13.75°. Follow through their AC from part (a).
Marks	**2**

WRITE IN THE ANSWER SPACE

$$\tan\theta = \frac{30}{122.5696\ldots} = 0.24476\ldots$$

$\theta = 13.8°$

Question 21 – Probability without replacement [9 marks]

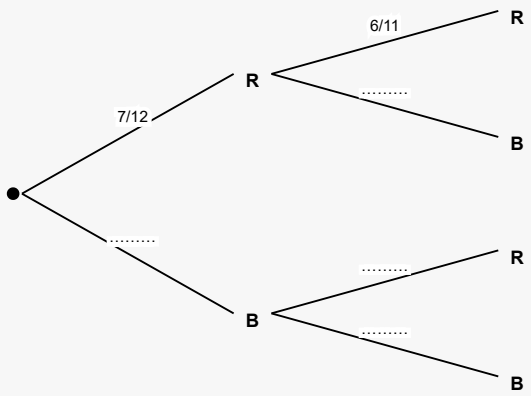
QUESTION 21 – MAIN PART

A bag contains 7 red counters and 5 blue counters.
Two counters are taken at random from the bag, without replacement.

21(a) [2 marks]

QUESTION

(a) Complete the tree diagram.



1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Probability} = \frac{\text{number of favourable counters}}{\text{total number of counters}}$$

Without replacement: the total goes down by 1 after each counter is taken

The probabilities on each pair of branches add up to 1

SUBSTITUTE THE VALUES

First counter (12 in the bag): P(R) = 7/12, P(B) = 5/12

After R (6 R and 5 B left, 11 in total): 6/11 and 5/11

After B (7 R and 4 B left, 11 in total): 7/11 and 4/11

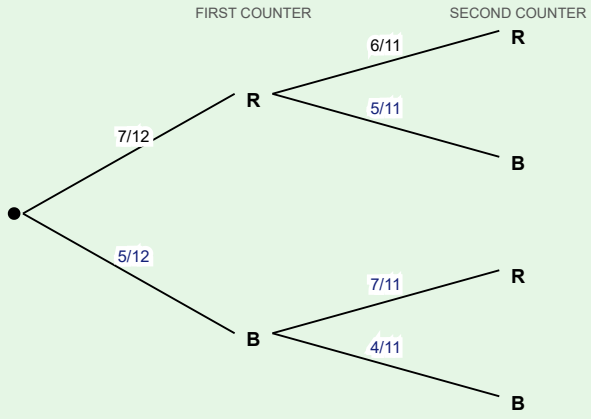
The counter is **not put back**, so only 11 are left for the second counter and the colour that was taken is one fewer. The two probabilities at each pair of branches must add up to 1.

2. MARK SCHEME

Answer	5/12 and 5/11 [B1] 7/11 and 4/11 [B1]
Marks	**2**

WRITE IN THE ANSWER SPACE

Complete the tree diagram (new values in blue):



21(b) [2 marks]

QUESTION

(b) Find the probability that both counters are red.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

P(A then B) = P(A) × P(B after A) (multiply along the branches)

SUBSTITUTE THE VALUES

$$P(R, R) = \frac{7}{12} \times \frac{6}{11} = \frac{42}{132}$$

Multiply **along** the top branch: 7/12 × 6/11 = 42/132 = 7/22.

2. MARK SCHEME

Answer	$\frac{7}{12} \times \frac{6}{11}$ [M1] $\frac{7}{22}$ [A1]
Guidance	Accept 42/132 or 0.318. Follow through from their tree.
Marks	**2**

WRITE IN THE ANSWER SPACE

$$P(R, R) = \frac{7}{12} \times \frac{6}{11} = \frac{42}{132}$$

$$\mathbf{\frac{7}{22}}$$

21(c) [2 marks]

QUESTION

(c) Find the probability that the two counters are different colours.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Different colours: P(R, B) + P(B, R) (add the two routes)

each route: P(X, Y) = P(X) × P(Y after X)

SUBSTITUTE THE VALUES

$$P(R, B) + P(B, R) = \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11}$$

There are **two ways**: red then blue, or blue then red. Multiply along each branch, then **add** the two answers: 35/132 + 35/132 = 70/132 = 35/66.

2. MARK SCHEME

Answer	$\frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11}$ [M1] $\frac{35}{66}$ [A1]
Guidance	Accept 70/132 or 0.530.
Marks	**2**

WRITE IN THE ANSWER SPACE

$$P(R, B) + P(B, R) = \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11}$$

$$= \frac{35}{132} + \frac{35}{132} = \frac{70}{132}$$

$$\mathbf{\frac{35}{66}}$$

21(d) [3 marks]

QUESTION

(d) Three counters are now taken at random from the original bag, without replacement. Find the probability that at least one of these counters is blue.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

P(at least one blue) = 1 − P(no blue)

P(R, R, R) = P(R) × P(R after R) × P(R after R, R)

SUBSTITUTE THE VALUES

$$P(R, R, R) = \frac{7}{12} \times \frac{6}{11} \times \frac{5}{10} = \frac{7}{44}$$

$$P(\text{at least one blue}) = 1 - \frac{7}{44}$$

“At least one” is always done with the **opposite**. The only way to have no blue is three reds: 7/12 × 6/11 × 5/10 = 7/44. So the answer is 1 − 7/44 = 37/44. Listing every way to get a blue would need seven products; the opposite needs one.

2. MARK SCHEME

Answer	1 − P(RRR) (the opposite idea) [M1] $\frac{7}{12} \times \frac{6}{11} \times \frac{5}{10}$ [M1] $\frac{37}{44}$ [A1]
Guidance	Accept 0.841 (3 s.f.). The last denominator must be 10.
Marks	**3**

WRITE IN THE ANSWER SPACE

$$P(RRR) = \frac{7}{12} \times \frac{6}{11} \times \frac{5}{10} = \frac{7}{44}$$

$$P(\text{at least one blue}) = 1 - \frac{7}{44}$$

$$\mathbf{\frac{37}{44}}$$

Question 22 – Grouped data [11 marks]

QUESTION 22 – MAIN PART

The table shows the times, t minutes, taken by 120 students to finish a test.

Time (t minutes)	$0 < t \leqslant 20$	$20 < t \leqslant 30$	$30 < t \leqslant 35$	$35 < t \leqslant 40$	$40 < t \leqslant 60$
Frequency	8	22	30	34	26

22(a) [4 marks]

QUESTION

(a) Calculate an estimate of the mean time.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Estimated mean $= \frac{\Sigma fx}{\Sigma f}$ (f = frequency, x = midpoint of the class)

Midpoint $= \frac{\text{lower boundary + upper boundary}}{2}$

SUBSTITUTE THE VALUES

Midpoints: 10, 25, 32.5, 37.5, 50

$\Sigma fx = 8\times10 + 22\times25 + 30\times32.5 + 34\times37.5 + 26\times50 = 4180$ and $\Sigma f = 120$

Mean $= \frac{4180}{120}$

For grouped data use the **midpoint** of each class. Multiply each midpoint by its frequency, add, then divide by the total frequency: 4180 ÷ 120 = 34.83…

2. MARK SCHEME

Answer	midpoints 10, 25, 32.5, 37.5, 50 [M1] frequency × midpoint, 80 + 550 + 975 + 1275 + 1300 [M1] 4180 ÷ 120 [M1] 34.8 [A1]
Guidance	Using class boundaries or class widths instead of midpoints scores 0.
Marks	**4**

WRITE IN THE ANSWER SPACE

Midpoints: 10, 25, 32.5, 37.5, 50

$\Sigma fx = 80 + 550 + 975 + 1275 + 1300 = 4180$

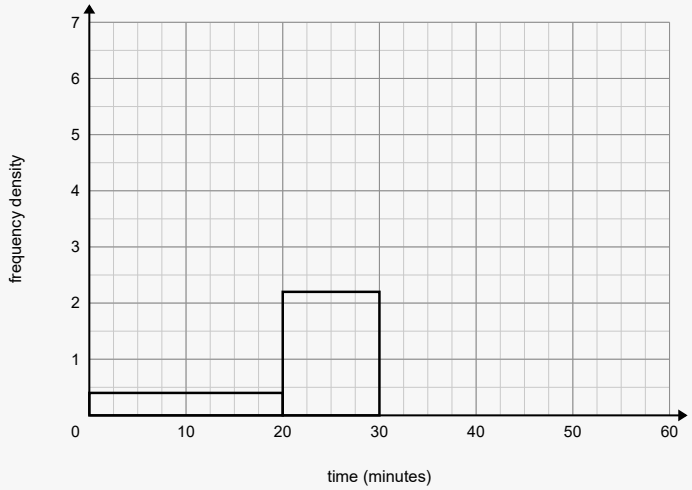
Mean $= \frac{4180}{120}$

34.8 minutes

22(b) [3 marks]

QUESTION

(b) Complete the histogram to show this information. The first two bars have been drawn for you.



1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

$$\text{Frequency density} = \frac{\text{frequency}}{\text{class width}} \quad \text{(the area of a bar = the frequency)}$$

SUBSTITUTE THE VALUES

$30 < t \leqslant 35$: $30 \div 5 = 6$ $35 < t \leqslant 40$: $34 \div 5 = 6.8$ $40 < t \leqslant 60$: $26 \div 20 = 1.3$

The class widths are unequal, so the height of each bar is the **frequency density**. On a histogram the **area** is the frequency, not the height – plotting 30, 34 and 26 scores zero. Check: 20(0.4) + 10(2.2) + 5(6) + 5(6.8) + 20(1.3) = 120 ✓.

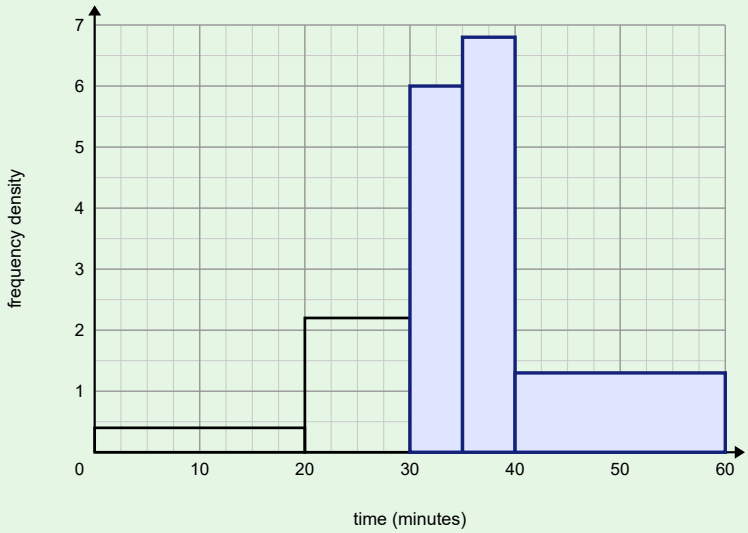
2. MARK SCHEME

Answer	frequency density = frequency ÷ class width [M1] bars of height 6, 6.8 and 1.3 drawn correctly [A2] (A1 if two bars are correct)
Marks	**3**

WRITE IN THE ANSWER SPACE

Frequency density: 30 ÷ 5 = 6, 34 ÷ 5 = 6.8, 26 ÷ 20 = 1.3

Draw the three bars:



22(c) [2 marks]

QUESTION

(c) Complete the cumulative frequency table and draw the cumulative frequency diagram on the grid.

Time (t minutes)	$t \leqslant 20$	$t \leqslant 30$	$t \leqslant 35$	$t \leqslant 40$	$t \leqslant 60$
Cumulative frequency	8				120

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Cumulative frequency = running total of the frequencies

Plot each total at the **upper** class boundary

SUBSTITUTE THE VALUES

8, 8 + 22 = 30, 30 + 30 = 60, 60 + 34 = 94, 94 + 26 = 120

Points: (20, 8) (30, 30) (35, 60) (40, 94) (60, 120)

Add up as you go. Plot each total at the **upper boundary** of its class – never in the middle. Start the curve at (0, 0) and join the points with a smooth curve.

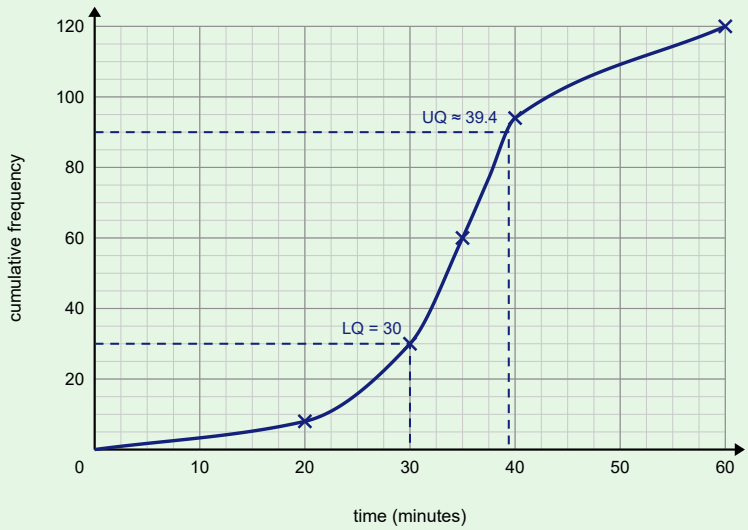
2. MARK SCHEME

Answer	30, 60, 94 in the table [B1] points plotted at the upper class boundaries and a smooth curve drawn [B1]
Marks	**2**

WRITE IN THE ANSWER SPACE

Table: 8, 30, 60, 94, 120

Plot and draw the curve:



22(d) [2 marks]

QUESTION

(d) Use your diagram to estimate the interquartile range.

1. TEACHER EXPLANATION

FORMULA (IN SYMBOLS)

Interquartile range = upper quartile (UQ) − lower quartile (LQ)

LQ is at ¼ × n; UQ is at ¾ × n (n = total frequency)

SUBSTITUTE THE VALUES

n = 120: LQ at ¼ × 120 = 30th value; UQ at ¾ × 120 = 90th value

LQ ≈ 30 and UQ ≈ 39.4, so IQR = 39.4 − 30

On a cumulative frequency graph use ***n***, not n + 1. Draw across from 30 and 90 to the curve and read down: lower quartile t ≈ 30, upper quartile t ≈ 39.4. IQR = 39.4 − 30 = 9.4.

2. MARK SCHEME

Answer	30 and 90 used to read the quartiles [M1] 9.4 [A1]
Guidance	Accept 9 to 10 from a correct curve. Follow through from their curve.
Marks	**2**

WRITE IN THE ANSWER SPACE

Lower quartile (30th value) = 30

Upper quartile (90th value) ≈ 39.4

IQR = 39.4 − 30

9.4 minutes